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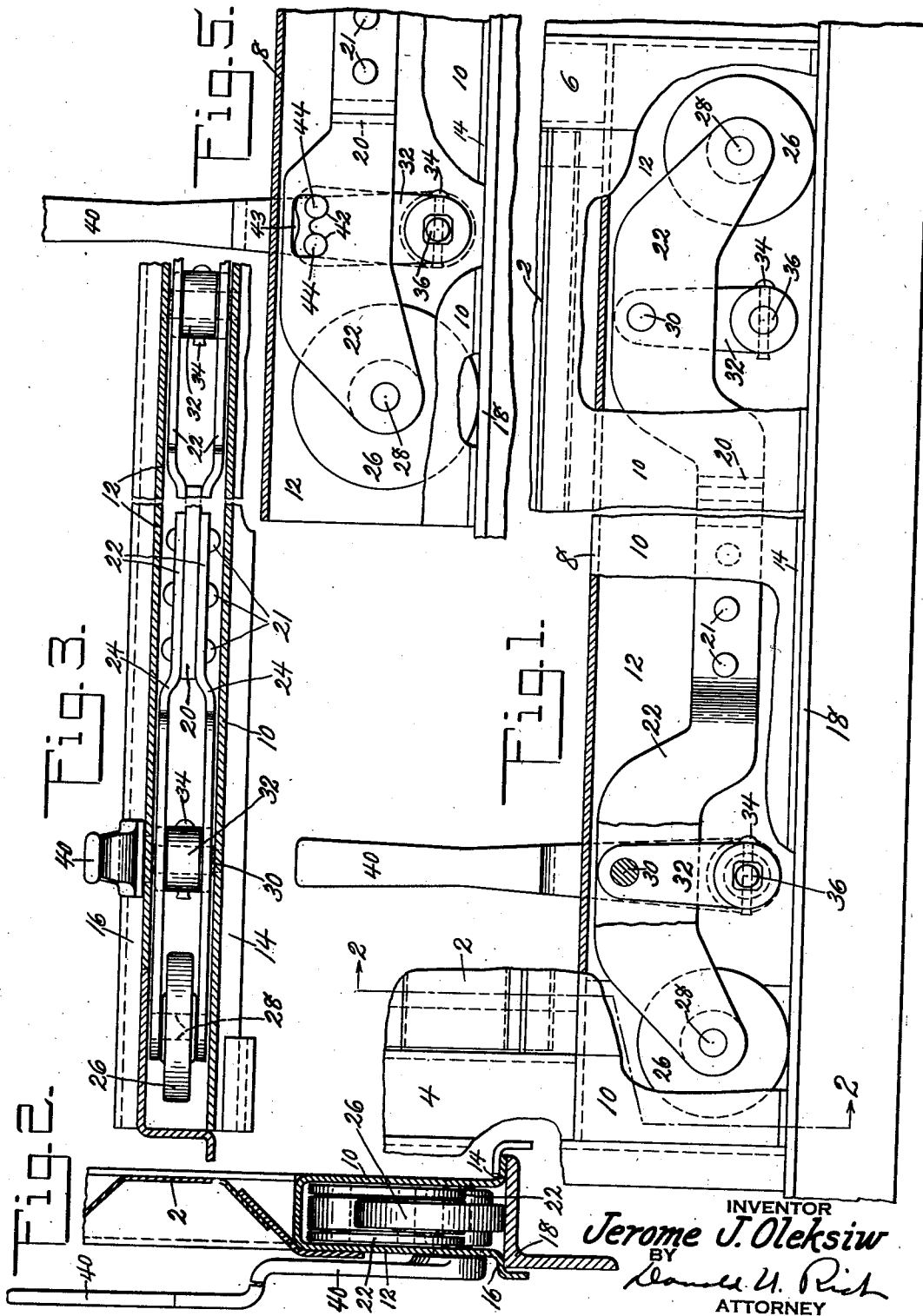
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2,263,468

VARIABLE LEVERAGE DOOR LIFT DEVICE

Filed Nov. 1, 1939

2 Sheets-Sheet 1



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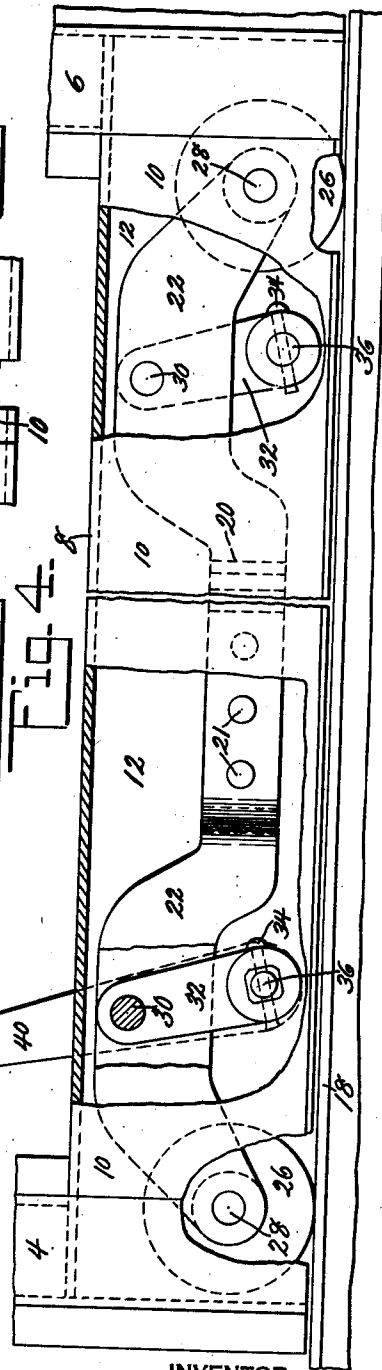
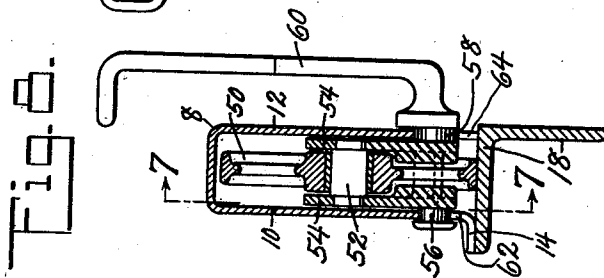
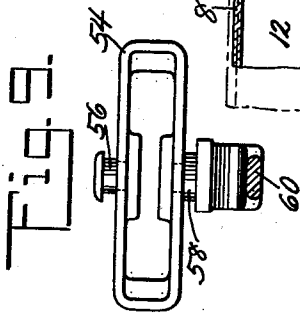
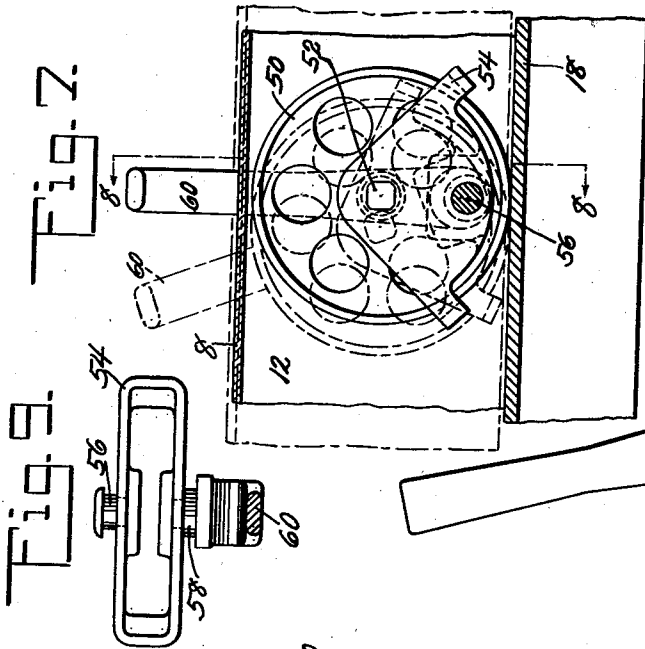
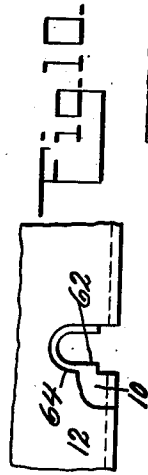
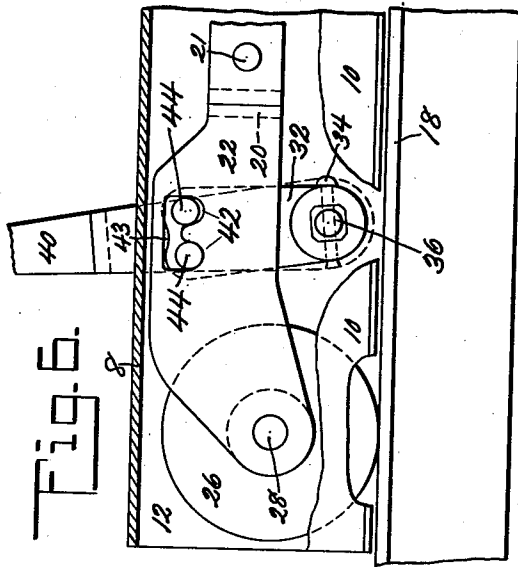
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VARIABLE LEVERAGE DOOR LIFT DEVICE

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2 Sheets-Sheet 2



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VARIABLE LEVERAGE DOOR LIFT DEVICE

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8 Claims. (Cl. 16—99)

This invention relates to doors in general and in particular to sliding doors which are normally held by frictional engagement with their track but may be raised on an antifriction device for horizontal sliding movement.

The majority of lift devices for sliding doors have had the force applied to the mechanism in a single direction regardless of the direction of door travel. A very few of the lift devices, however, have been made in which the force as applied to lift the door was in the direction of door movement, but in any case a definite fixed leverage was utilized. This fixed leverage was sufficient to enable the average operator to lift the door under average conditions, but in case the door was frozen in place or the parts had become rusted, then it became impossible for the average operator to lift the door. It is an object, therefore, of the present invention to provide a door lift device with the parts so arranged as to give a constantly varying leverage.

A further object of the invention is the provision of an improved roller lift device in which the leverage varies from an infinite value to a finite value, limited only by the ratio of the lengths of the links use.

A still further object of the invention is the provision of a variable leverage door lift device in which the force applied to the handle to lift the door is always in the same direction as that toward which the door is to be slid and will at some point be equal to the force necessary to cause the sliding movement of the door.

Yet another object of the invention is the provision of a variable leverage door lift device in which the lowering motion of the raised door causes the door to rock toward the direction it has been moved, that is, to rock toward closed and sealed position during lowering and after the door has been slid into substantially closed position.

These and other objects of the invention will be apparent to persons skilled in the art from a study of the following description and accompanying drawings, in which:

Figure 1 is an elevational view of the lower portion of the door with parts broken away to more clearly disclose the construction;

Fig. 2 is a sectional view taken substantially on line 2—2 of Figure 1;

Fig. 3 is a plan view of the lift mechanism with the door housing being shown in section;

Fig. 4 is a view similar to Figure 1 but showing the door in a raised position;

Fig. 5 is a partial elevational view similar to

Figure 1 but showing a modification of the construction;

Fig. 6 is an elevational view of the construction shown in Fig. 5 but with the parts in door raised position;

Fig. 7 is an elevational sectional view taken substantially on line 7—7 of Fig. 8 and showing a still further modification;

Fig. 8 is a sectional view taken substantially on line 8—8 of Fig. 7 but with the roller housing incomplete;

Fig. 9 is a plan view of the roller carriage used in the construction of Figs. 7 and 8, and

Fig. 10 is an elevational view of a small portion of the door bottom rail disclosing the manner in which the door is mounted on the roller carriage construction of Figs. 7, 8 and 9.

Referring now to the drawings in detail, it will be seen that the mechanism has been applied to a more or less conventional freight car door having a corrugated central panel 2 welded or otherwise secured to front stile 4, rear stile 6 and to bottom rail 8. In the form illustrated this bottom rail is of downwardly facing channel form having its inner and outer flanges or legs 10 and 12 terminating in inner and outer re-flanges 14 and 16 respectively adapted to normally rest in frictional engagement with an angular shaped track 18. It is, of course, obvious that any type of door may be used and that any type of bottom rail may also be used, but the type shown is preferable since it fully houses the lift mechanism and prevents entrance of foreign matter.

The door as shown in Figures 1 to 6 inclusive is mounted upon a roller carriage consisting of an intermediate connecting bar 20 joined at its ends by rivets or other means 21 to straps 22 offset as at 24 and extended to receive therebetween a roller 26 rotatably mounted upon axle 28. The straps 22 are preferably offset upwardly intermediate their ends in order that the bar 20 may be located as low as possible, thus lowering the center of gravity of the carriage and at the same time providing clearances for the motion of the mechanism. Each strap 22 is pierced intermediate its ends in order to receive pin 30 extending between the straps and rotatably carrying thereon a link 32 as clearly shown in Figures 1 to 4 inclusive. The lower end of this link is non-rotatably connected by means of pin or rivet 34 to a transverse pin 36 journaled at its ends in the downwardly directed flanges 10 and 12 of the bottom rail. One of these latter transverse pins may be made square as clearly shown

in the left hand portion of Figures 1 and 4 and is extended outwardly beyond the flange 12 of the bottom rail and has attached rigidly thereto an operating handle 40. This construction may be readily assembled by inserting the carriage complete with links into the bottom rail, after which the pins 36 are inserted and fastened by the pin or rivet 34 and the operating handle secured to the transverse shaft which extends outwardly beyond the bottom rail.

The operation of the construction shown in Figures 1 to 4 is believed obvious, but assuming the door to be in lowered position of Figure 1 it may be raised to the position shown in Figure 4 by merely applying a force to the handle 40 in the direction toward which the door is to be moved. It should be noted that with the handle 40 vertical and at the start of the lifting motion the leverage is infinite and constantly varies as the handle is swung, that is, as the handle 40 starts its swing the motion of the pins 36 is tangent to the track and as it reaches the end of its travel the pins are moving substantially perpendicular to the track. In other words, the effective lever arm between the fulcrum 30 and point of weight application 36 varies from zero to the length of link 32 if this link is moved to a horizontal position, while the length of the lever arm between the point of force application and the fulcrum remains constant.

In other words, the leverage will vary from an infinite to a finite value limited by the ratio of the lengths of the links used. In general practice, however it is not necessary to provide for such an extreme lift of the door and accordingly the carriage may be designed as illustrated and in this case the minimum finite leverage is limited by the swing obtainable before pin 36 strikes the straps 22. It will be seen from Figure 4 that as the door is moved toward a closed position and then lowered that the door will rock forward toward a further closed position during the lowering of the door; such a motion being due to the relation of the parts and to the fact that the friction of the door on its pins is less than the friction between the carriage and track.

In the form shown by Figures 5 and 6 the major portions are identical with that just described in connection with Figures 1 to 4 and accordingly the same reference numerals will be used. In this form, however, two holes 42 are made in each strap and joined by an opening 43, thus providing a window like opening having two bearing portions in the lower surface thereof. The links 32 are also provided with two holes adapted to receive two transverse pins 44. Except for the change in the mounting of the links 32 this modification is exactly the same as that previously described and will be assembled and operated in the same manner. However, there is this difference, the leverage varies from a small finite value to an increased finite value dependent upon the ratio of the lengths of the links or upon the swing permitted the pin 36. The value of the lower finite leverage will be dependent upon the distance the pins 44 are placed on either side of the center line through the links. By this arrangement of offset pins a greater throw or lift may be obtained without necessitating a great change in the anti-friction carriage, for the initial rotation starts about a pin offset from the center line, not about a pin on the center line.

In the form shown in Figures 7 to 10 inclusive an independent roller 50 is used journaled on a transverse pin 52 resting at its ends in the side

walls of a roller carriage 54. The inner wall of this roller carriage is formed with a projecting trunnion 56, while the outer wall is also provided with a projecting trunnion 58 formed integral with an operating handle 60. It should be noted that the trunnions 56 and 58 are offset relative to the roller axis or axle and accordingly any rocking motion given the handle will tend to rotate the axle about the trunnions 56 and 58. The entire unit may be readily inserted in the bottom rail of the door by lifting the unit into position with trunnions 56 and 58 passing into slots 62 and 64 formed respectively in the inner and outer flanges of the door bottom rail. Application of force to the handles in the direction of desired door movement will automatically raise the door and no conscious lifting effort may be exerted by the operator.

In each of the forms described application of force to the operating handle in the direction of desired door movement automatically applies a powerful leverage to lift the door and this leverage varies constantly decreasing with increased door lift. It will be obvious that regardless of the proportioning of the parts there will be some point at which the force applied to the handle will be sufficient to cause the door to roll on the track and this point will only be reached after the door has been lifted a slight amount on the carriage or roller for, as previously stated, the leverage decreases either from an infinite or extremely small finite value. By making the leverage variable any operator will be able to exert an extremely powerful lifting leverage on the door to break the seal between the door and car and to break any ice or other matter that may be preventing the lift of the door.

While the constructions shown have been described more or less in detail, it will be obvious that various modifications, rearrangements of parts and changes in construction will suggest themselves to persons skilled in the art and all such modifications, rearrangements of parts and changes in construction are contemplated as will fall within the scope of the appended claims which define my invention.

What is claimed is:

1. In a lift device for sliding doors having the bottom portion normally resting on a supporting track, the combination of an anti-friction carriage, suspension links hanging from said carriage and connected adjacent their lower ends to the lower portion of the door thereby suspending the door on said carriage, and an operating handle connected to said suspension links to shift the same, and lift the door.

2. In a lift device for sliding doors having the bottom portion normally resting on a supporting track, the combination of an anti-friction device including a frame and spaced rollers, suspension links hanging from said frame and connected adjacent their lower ends to the lower portion of the door thereby suspending the door on said frame, and an operating handle connected to one of said suspension links to shift the same, and lift the door, said handle and suspension links being normally positioned substantially in a vertical plane common to both whereby a variable leverage may be exerted on the door to lift the same during shifting of the handle.

3. In a lift device for sliding doors having a bottom portion normally engaging a supporting track, the combination of an anti-friction carriage, suspension links rockably supported ad-

adjacent their upper ends on said carriage, means supporting said door in stable equilibrium on said links adjacent the lower ends thereof, and an operating handle secured to at least one of said last named means.

4. In a lift device for sliding doors having a bottom portion normally engaging a supporting track, the combination of a carriage mounted on anti-friction rollers, means suspended on said carriage and normally supporting said door at points beneath the carriage in stable equilibrium, and an operating handle rigidly connected to said suspension means.

5. In a lift device for sliding doors having a bottom portion normally resting on a supporting track, the combination of a carriage mounted on anti-friction rollers, suspension means connected to said carriage and supporting said door at points beneath the carriage, and an operating handle rigidly connected to said suspension means, said suspension means being connected to the carriage by spaced trunnions located on each side of a vertical plane through the door support points.

6. In a lift device for sliding doors having a bottom portion normally resting on a supporting track, the combination of a carriage mounted adjacent its ends on rollers, a plurality of suspension links trunnioned on said carriage intermediate said rollers, means connecting the lower ends of said links and door, and an operating handle rigidly connected to at least one of said

last named means and the adjacent link to rock the latter about its carriage trunnion.

7. In a lift device for sliding doors having a bottom portion normally resting on a supporting track, the combination of a carriage mounted adjacent its ends on rollers, a plurality of suspension links trunnioned on said carriage intermediate said rollers, means connecting the lower ends of said links and door, and an operating handle rigidly connected to at least one of said last named means and the adjacent link to rock the latter about its carriage trunnion, said last named means and trunnion for each link being located substantially in a vertical plane containing the trunnion axis.

8. In a lifting device for sliding doors having a bottom portion normally engaging a track, the combination of an anti-friction carriage, suspension links hanging from said carriage and connected adjacent their lower ends to the lower portion of the door thereby suspending the door on said carriage, an operating lever connected to the lower portion of one of said suspension links to shift the same, said lever extending in the same direction as the link to which it is connected and the relation of said link and lever being such that initial predetermined lift applied to said lever will result in disengaging said door from said track and further movement of said lever will cause said door to move in the direction of the lever movement.

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